Lab Evaluation:

Course Name: Data Structures & Algorithm	**Course Code:** CMPE-212L
Assignment Type: Lab	**Dated:** 26th March 2024
Semester: 4th	**Session:** 2022
Lab/Project/Assignment #: 9	**CLOs to be covered:** CLO 2
Lab Title: Hash Table	**Teacher Name:** Engr. Afeef Obaid

CLO 2	Design solutions for problems using appropriate data structures and algorithms					
Levels (Marks)	**Level 1**	**Level 2**	**Level 3**	**Level 4**	**Level 5**	**Level 6**
(10)						
					Total	**/10**

Rubrics for Current Lab Evaluation

Scale	Marks	Level	Rubric
Excellent	**9-10**	L1	Submitted all lab tasks, BONUS task, have good understanding.
Very Good	**7-8**	L2	Submitted the lab tasks but have good understanding
Good	**5-6**	L3	Submitted the lab tasks but have weak understanding.
Basic	**3-4**	L4	Submitted the lab tasks but have no understanding.
Barely Acceptable	**1-2**	L5	Submitted only one lab task.
Not Acceptable	**0**	L6	Did not attempt

LAB No. 9

Lab Goals/Objectives:

- Implementation of string-based Hash table using Open Addressing

Equipment Required: Computer and Dev C++ Compiler

Theory/Relevant Material: Hash Tables, Open Addressing, Secondary Hashing

Lab Tasks:

- Complete the definition of the missing function in the code provided of Hash Table. Incomplete functions:
- Search // search given key value in the given hash table
- Insert // insert given key into hash table
- HFQuad1 // hash function to provide hash value based on the string key
- HFQuad2 // hash function to provide hash value based on the string key

Run some experiments to see:

a) How the average number of probes per insertion differs when using different hash functions.

b) A critical influence on performance of an open addressing hash table is the load factor; that is, the proportion of the slots in the array that are used. As the load factor increases towards 100%, the number of probes that may be required to find or insert a given key rises dramatically. Once the table becomes full, probing algorithms may even fail to terminate. Even with good hash functions, load factors are normally limited to 80%. A poor hash function can exhibit poor performance even at very low load factors by generating significant clustering, especially with the simplest linear addressing method.

Experiment with load factor 50% (engaging the size of hash table double as compared to the values to be stored), 60% and 70%. Does changing the load factor effects the number of probes with each hash function. of open addressing. Under what factors quadratic probing performs best?